ECE 329 **Homework 10** **Due: Mon, Dec 8, 2025, 11:59PM**

4 problems totaling 50pts.

1. (10pts) A y-polarized plane TEM wave is propagating in vacuum (i.e., $v = c \approx 3 \times 10^8$ m/s and $\eta = \eta_o \approx 120\pi\,\Omega$) in $+z$ direction. The time variation of the wave field at $z = 0$ is described as $E_y(0,t) = 4\triangle(\frac{t}{\tau})$ V/m, where $\tau = 4\,\mu$s and $\triangle(\frac{t}{\tau})$ is the unit triangle function with width τ (i.e., non-zero for $-\tau/2 \le t \le \tau/2$).

   a) (2pts) Determine the vector wavefield $\mathbf{E}(z,t)$ written explicitly in terms of space and time variables z and t and the speed of light $c \approx 3 \times 10^8$ m/s in vacuum.

   b) (2pts) Determine the accompanying wavefield $\mathbf{H}(z,t)$ in A/m units written in terms of z and t, the speed of light c in vacuum, and the intrinsic impedance $\eta = \eta_o \approx 120\pi\,\Omega$ for vacuum. Remember that the magnitude of $\mathbf{H}$ equals magnitude of $\mathbf{E}$ divided by η and direction of $\mathbf{H}$ is perpendicular to $\mathbf{E}$ such that vector $\mathbf{E}\times\mathbf{H}$ (known as *Poynting vector*) is pointed in the direction of propagation ($+z$ in this case).

   c) (2pts) Find the expression for $E_y(z,t)$ at $z = +1500$ m and plot it vs t.

   d) (2pts) Find the expression for $E_y(z,t)$ at $t = 12\,\mu$s and plot it vs z.

   e) (2pts) Find the expression for $H_x(z,t)$ at $t = 12\,\mu$s and plot it vs z.

2. (10pts) In a homogeneous ideal dielectric with $\epsilon = \epsilon_r \epsilon_o$ and $\mu = \mu_r \mu_o$ a plane TEM wave with the following components is observed:

$$\mathbf{E} = \hat{x} u(t - \frac{z}{c/2}) + \hat{y} g(t - \frac{z}{c/2}) \, \frac{\mathrm{V}}{\mathrm{m}}$$

and

$$\mathbf{H} = \hat{x} \frac{10z}{c} - \hat{x} 5t + \hat{y} \frac{1}{120\pi} u(t - \frac{2z}{c}) \, \frac{\mathrm{A}}{\mathrm{m}},$$

where $u(t)$ denotes the unit-step function and c is the speed of light in free space. Using the above information,

a) (2.5pts) Determine the intrinsic impedance η for the medium.

b) (2.5pts) Determine the propagation velocity v.

c) (2.5pts) Determine ϵ_r and μ_r.

d) (2.5pts) Function $g(t)$.

3. (10pts) For each of the four plane waves (in free space) shown below, wave fields **E** and **H** have the d'Alembert form $\propto f(t \mp \frac{s}{c})$, where s is some coordinate (x, y, or z) and $f(t)$ function is either $\cos(\omega t + \phi)$ or $\sin(\omega t + \phi)$ and $\beta = \omega/c$, a parameter known as wavenumber, and ϕ some "phase shift".

   a) (2.5pts) $\mathbf{E}_1 = 4\sin(\omega t - \beta z)\hat{x}\,\mathrm{V/m}$

   b) (2.5pts) $\mathbf{E}_2 = 5\cos(\omega t - \beta z)\hat{x} - 5\cos(\omega t - \beta z)\hat{y}\ \mathrm{V/m}$

   c) (2.5pts) $\mathbf{H}_3 = 2\cos(\omega t + \beta z + \frac{\pi}{3})\hat{x} - 2\cos(\omega t + \beta z - \frac{\pi}{6})\hat{y}\,\mathrm{A/m}$

   d) (2.5pts) $\mathbf{H}_4 = 4\cos(\omega t + \beta x)\hat{z} + 4\sin(\omega t + \beta x)\hat{y}\,\mathrm{A/m}$

   For each of the four plane waves shown above, determine the expression for **H** or **E** that accompanies the given wave field.

4. (20pts) We have on $x = 0$ plane a pulse of sheet current $\mathbf{J}_s(t) = \hat{y}\, At\ \mathrm{rect}(\frac{t}{\tau})$ A/m, where $\tau = 2\,\mu$s and $A = \frac{4}{\mu s}$. Regions adjacent to the current sheet are vacuum. The current pulse generates traveling waves of $\mathbf{E}(\mathbf{r}, t) = \hat{y}E_y(x, t)$ and $\mathbf{H}(\mathbf{r}, t) = \hat{z}H_z(x, t)$.

   a) (2pts) Plot $J_{sy}(t)$ vs t over $-3 < t < 3\,\mu$s.

   b) (4pts) Plot $E_y(x, t)$ vs t at locations $x = 0$ and $x = \pm 300$ m.

   c) (4pts) Plot $H_z(x, t)$ vs t at locations $x = 0^{\mp}$ and $x = \pm 300$ m.

   d) (4pts) Determine the TEM wave energy radiated per unit area (in J/m^2 units) by the current pulse $\mathbf{J}_s(t)$.

      Hint: integrate the power injected per unit area, $-\mathbf{J}_s \cdot \mathbf{E}$, over the duration of pulse $\mathbf{J}_s(t)$.

   e) (4pts) Determine the TEM wave energy transported by the radiated fields across each unit area (in J/m^2 units) on the $x = 300$ m plane.

      Hint: integrate the Poynting vector $\mathbf{E} \times \mathbf{H}$, over the durations of the outgoing $\mathbf{E}(300\hat{x}, t)$ and $\mathbf{H}(300\hat{x}, t)$ pulses produced by $\mathbf{J}_s(t)$.

   f) (2pts) Are the numbers found in parts (d) and (e) compatible with the principle of energy conservation? Discuss.